Privacy-Preserving Federated Learning for IoT-Based Healthcare

A Layered-Defense Benchmark on ECG Arrhythmia Classification

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The problem

- Wearable health devices stream clinically valuable but **highly sensitive** ECG data.
- Centralised training is the default $\Rightarrow$ single point of failure, re-identification risk, HIPAA/GDPR friction.
- "Privacy-preserving FL" = DP-SGD + secure aggregation + compression. **Textbook recipe**.
- The textbook does *not* say **which layer pays for which threat**, or **what each costs** on imbalanced clinical data.

Gap addressed

Empirical / measurement gap, not a new primitive. Build the layered-defence framework, measure the marginal cost of each layer on a realistic healthcare workload, surface what the headline-accuracy convention hides.

Contributions

C1 **A layered-defence FL framework** with three independently switchable privacy layers (DP-SGD, pairwise-mask SecAgg, top-k + int8 compression). $\sim$2,500 lines of Python; runs end-to-end in under 30 minutes on a single GPU.

C2 **Cryptographic correctness as a code-level invariant.** The SecAgg cancellation property is encoded as a unit test that runs in CI, not just argued in prose.

C3 **A class-prior collapse diagnostic for DP on imbalanced healthcare data.** Final test accuracy at $\varepsilon = 5$ equals class-N prevalence to four decimals — the model is the majority-class predictor.

The privacy stack

Three layers, each addressing a different threat:

Layer	**Mechanism**	**Threat**
Local training	DP-SGD (Opacus, RDP accountant)	Membership inference; gradient leakage
Aggregation	Pairwise-mask SecAgg	Honest-but-curious server; per-client tracing
Communication	Top-k sparsification + int8 quant.	Bandwidth budget for IoT

Three orthogonal switches in code: `dp_config`, `secagg_session`, `compression_method`. Each can be `None`; the corresponding layer degenerates to identity. This is what makes per-layer cost measurement possible.

Threat model: five adversaries

	Capability	**Goal**	**Mitigation**
A1 Honest-but-curious server	Sees every per-client message	Per-client membership inference	SecAgg
A2 Coalition of clients ($t < N$)	Pool masks	Unmask non-coalition update	SecAgg (pairwise)
A3 Network eavesdropper	Wire observation	Read updates	TLS + SecAgg
A4 Adversary holding w_T	Inspect / query released model	Membership inference at the model	DP-SGD
A5 Byzantine client	Submit crafted updates	Poison or backdoor	Out of scope

Trust assumptions: TLS in place, secure PRG, no more than $N - 2$ clients collude, no hardware side channels.

Headline results — records 100–115 (97% N)

Configuration	Test acc	Final ε
Centralized (5 epochs)	**0.9878**	—
FedAvg, 4 clients, 10 rounds	**0.9880**	—
FedAvg + SecAgg	0.9860	—
FedAvg + DP, target $\varepsilon = 5$, 20 rounds	0.9711	4.9955
Full pipeline (DP+SecAgg+Comp), $\varepsilon = 8$, 30 rounds	0.9711	7.9964
Class-*N* prevalence (predict-all-N)	**0.9716**	—

Three rows, three findings. The bottom row is the clue.

Finding 1 — Federation is free

- Centralized: **0.9878**
- FedAvg (10 rounds): **0.9880**
- Gap: **0.02 pp** — well below seed variance.

No proximal regulariser, no server momentum, no per-client correction. Plain FedAvg suffices when the model is not over-parameterised and client distributions are not adversarially heterogeneous. This is the empirical baseline against which every subsequent privacy layer is measured.

Round-by-round, vanilla FedAvg

0.9711, 0.9711, 0.9758, 0.9840, 0.9840, 0.9724, 0.9831, 0.9864, 0.9862, **0.9880**.

First two rounds match class-N prevalence; round 3 onwards picks up the V-class signal.

Finding 2 — SecAgg is free + correctness in CI

- FedAvg + SecAgg: **0.9860**
- Gap vs. FedAvg: **0.20 pp** (float32 mask-cancellation noise).
- Modular-integer arithmetic would close it; unnecessary for a simulation.

C2 contribution: the cancellation argument is encoded as one assertion in CI:

The cryptographic correctness invariant

`tests/test_secagg.py::test_masks_cancel_under_aggregation`

Sum of $N = 4$ masked updates equals sum of unmasked updates to 10^{-4} in ℓ_2.

Pairwise mask cancellation:

$$\tilde{\Delta w}_i = \Delta w_i + \sum_{j>i} m_{ij} - \sum_{j<i} m_{ji}$$

$$\sum_{i=1}^{N} \tilde{\Delta w}_i = \sum_{i=1}^{N} \Delta w_i.$$

Every m_{ij} appears once with $+$ and once with $-$. Exact cancellation. *Recommendation:* run SecAgg by default — below-noise utility cost, real threat coverage.

Finding 3 — DP-SGD collapses to majority-class

Test accuracy round by round, DP-SGD ($\varepsilon \approx 5$):

r1	r2	r3	r4	...	r20
0.9711	0.9711	0.9711	0.9711	...	0.9711

Class-N prevalence: $4{,}934/5{,}078 = 0.9716$.
Final DP-SGD accuracy: 0.9711.
Agreement to **four decimals**.

The collapse, in one line

The model is, with statistical certainty, the *majority-class predictor*.

Train accuracy still rises $0.9147 \rightarrow 0.9712$ across the same 20 rounds. The model learns *something* during training; that something is "predict N for everything." Adding compression on top does not degrade further at $\varepsilon = 8$: same 0.9711.

Why this matters — the headline-accuracy trap

Casual reading of the table

DP at $\varepsilon = 5$ scores 0.9711 vs. FedAvg's 0.9880.
"DP costs about 1.7 pp." Publishable.

Honest reading

DP costs the model's ability to detect any rare class *at all*.
The 0.9711 hides the failure precisely because the metric matches the cliff.

The thesis-level recommendation. Any privacy-preserving FL paper for imbalanced healthcare data must report **majority-class baseline** and **macro-F1** alongside top-1 accuracy. The privacy/utility tradeoff curve is misleading without them.

Follow-up — records 200–234 (82% N)

Same five experiments. **No hyperparameters retuned.** Different MIT-BIH slice with moderate (not extreme) imbalance.

Configuration	100–115 acc	200–234 acc	vs. majority
Centralized (5 epochs)	0.9878	0.9389	+11.81 pp
FedAvg (10 rounds)	0.9880	0.9636	+14.28 pp
FedAvg + SecAgg (10 rounds)	0.9860	0.9558	+13.50 pp
FedAvg + DP, $\varepsilon \approx 5$ (20 rounds)	0.9711	**0.8845**	**+6.37 pp**
Full pipeline, $\varepsilon \approx 8$ (30 rounds)	0.9711	0.8176	−0.32 pp

- DP-only **climbs out** of the majority plateau — the cliff is a property of *extreme* imbalance, not DP-SGD per se.
- Full pipeline still collapses — top-k+int8 on top of DP noise pushes the regime back across the cliff.

Practical recommendations for deployment

1. **Run SecAgg by default.** Below-noise utility cost, threat coverage for A1–A3, correctness in one CI test.
2. **Treat DP-SGD on imbalanced clinical data as non-trivial.** Always report majority-class baseline and macro-F1.
3. **Compression is free in the post-DP regime.** If the model is at the privacy floor, top-k+int8 buys $\sim$40$\times$ bandwidth at no further utility cost. If the model is not at the floor, compression introduces real approximation error and must be measured.
4. **Benchmark on your actual class distribution** before fixing the budget. The cliff exists; its location depends on the workload.

Limitations (be candid)

- **Single-seed** runs — multi-seed across all five configurations is the first item of future work.
- **Simulation only** — Flower in-process, no real network or device-compute constraints.
- **SecAgg is a prototype** — pre-shared seeds (no Diffie–Hellman), no Shamir secret-sharing (no dropout tolerance), float32 not modular integer.
- **Per-record DP, not per-patient.** Group privacy gives a vacuous bound at the patient level for our partition.
- **No comparison to FedProx / SCAFFOLD / Krum** — FedAvg already matches centralized; out of scope.
- **Macro-F1 not logged round-by-round** for the federated runs — would have surfaced the collapse earlier.

Future work — what fits cleanly on top

1. **Multi-seed runs** across all five configurations; error bars on the headline table.
2. **Class-weighted DP-SGD** on records 200–234 (where DP-SGD alone clears the cliff and the full pipeline does not).
3. **Modular-integer SecAgg** to remove the 0.20 pp float32 cancellation noise.
4. **Byzantine-robust aggregation** (Krum, median, trimmed mean) composed with the existing privacy stack.
5. **MIMIC-IV vital-signs port** to test whether class-prior collapse generalises beyond ECG.
6. **Diffie–Hellman + Shamir secret-sharing** to upgrade the SecAgg prototype to a deployment-grade, dropout-tolerant protocol.

Conclusion

- Built a layered-defence FL framework with three independently switchable privacy layers on top of FedAvg. Five reproducible experiments; every number tied to one YAML config.
- **Federation is free** on this workload (0.02 pp gap).
- **SecAgg is free** (0.20 pp), and its cryptographic correctness is a one-line CI invariant.
- **DP-SGD collapses to majority-class** on extreme imbalance — a privacy/utility cliff that the headline-accuracy convention hides. Dataset-conditional: on moderate imbalance DP-SGD alone clears the cliff but the full pipeline still does not.
- **One recommendation for the field:** report majority-class baseline and macro-F1 alongside top-1 accuracy. Always.

Thank you

Questions?

Code, configs, thesis source:
`https://github.com/mapeke/privacy-preserving-fl-healthcare-iot`